Hints: Limits – 2. Recurrences solvable in closed form

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Let $f(x) = (x^2 - 9)/8$. Its fixed points are 9 and -1; note $f'(9) = \frac{9}{4}$ (repelling) and $f'(-1) = -\frac{1}{4}$ (attracting). Also f is even and $f(x) \geqslant -\frac{9}{8}$, so only $|x_1|$ matters and $x_n \geqslant -\frac{9}{8}$ for $n \geqslant 2$.

Hint 2. Use $f(x) - 9 = \frac{(x-9)(x+9)}{8}$: if $|x_1| > 9$ then $x_2 > 9$ and $x_n - 9$ grows at least geometrically, so $x_n \to +\infty$; if $|x_1| = 9$ then $x_n = 9$ for $n \geqslant 2$.

Hint 3. If $|x_1| < 9$ then $x_2 \in [-\frac{9}{8}, 9)$. Use $f(x) + 1 = \frac{(x+1)(x-1)}{8}$: for $x \in [-\frac{9}{8}, 9)$ we have $|x - 1| < 8$, so $d_n = |x_n + 1|$ decreases; then all later x_n lie in $[-\frac{9}{8}, -1 + d_2]$, where $|x - 1| \leqslant 8q$ for a fixed $q < 1$, so $d_n \to 0$ geometrically.

Hint 4. Answer: $\lim x_n = -1$ if $|x_1| < 9$, $= 9$ if $|x_1| = 9$, $= +\infty$ if $|x_1| > 9$.

Problem 2

Hint 1. Uniqueness and existence: $g(x) = x - \varepsilon \sin x$ is continuous with $g'(x) = 1 - \varepsilon \cos x \geqslant 1 - |\varepsilon| > 0$, and $g(x) \to \pm\infty$ as $x \to \pm\infty$.

Hint 2. Rewrite the equation as $x = t + \varepsilon \sin x$ and iterate $x_{n+1} = t + \varepsilon \sin x_n$ from any x_0. Since $|\sin u - \sin v| \leqslant |u - v|$, the map is a contraction with constant $|\varepsilon| < 1$: $|x_{n+1} - x| \leqslant |\varepsilon|\,|x_n - x|$, so $x_n \to x$ geometrically (this is Kepler's equation).

Problem 3

Hint 1. Solve the recurrence: the characteristic roots are 2 and $\frac{1}{2}$; find $F(n)$ in closed form.

Hint 2. $F(n) = 2^n - 2^{-n}$, so with $y_n = 2^{2^n}$ the terms are $\dfrac{1}{y_n - 1/y_n} = \dfrac{y_n}{y_n^2 - 1}$, and $y_{n+1} = y_n^2$.

Hint 3. $\dfrac{y}{y^2 - 1} = \dfrac{1}{y - 1} - \dfrac{1}{y^2 - 1}$, so the series telescopes: $\sum_{n \geqslant 0} \frac{1}{F(2^n)} = \frac{1}{y_0 - 1} = 1$. Rational.

Problem 4

Hint 1. Compute: $1, 2, 6, 22, 90, 394, \ldots$ These are the (large) Schröder numbers, which satisfy the integral recurrence $s_n = s_{n-1} + \sum_{k=0}^{n-1} s_k s_{n-1-k}$, i.e. their generating function $S(x)$ satisfies $S = 1 + xS + xS^2$. The task is to prove $a_n = s_n$.

Hint 2. Translate the given recurrence into an equation for $f(x) = \sum a_n x^n$: multiplying by x^{n+2} and summing gives the first-order linear ODE

$$x(1 - 6x + x^2) f'(x) + (1 - 3x) f(x) = 1 + x.$$

A formal power series solution of this ODE is unique (the coefficient of x^m determines a_m), so it suffices to check that S satisfies it.

Hint 3. From $S = 1 + xS + xS^2$ get $S(x) = \dfrac{1 - x - \sqrt{1 - 6x + x^2}}{2x}$; differentiate and verify the ODE. Then $a_n = s_n$, which are integers by the convolution recurrence.

Problem 5

Hint 1. Use the generating function $f(x) = \sum a_n x^n$ (by induction $0 < a_n \leqslant 1$, so it converges for $|x| < 1$). The required limit is $f(\frac{1}{2})$.

Hint 2. The recurrence says $(n+1)a_{n+1} = \sum_{k=0}^{n} a_k \cdot \frac{1}{n-k+2}$, i.e. $f'(x) = f(x)g(x)$ with $g(x) = \sum_{m\geqslant 0} \frac{x^m}{m+2} = \dfrac{-\ln(1-x) - x}{x^2}$.

Hint 3. An antiderivative of g vanishing at 0 is $1 + \frac{(1-x)\ln(1-x)}{x}$, so $f(x) = e\,(1-x)^{(1-x)/x}$ and the limit is $f(\frac{1}{2}) = \dfrac{e}{2}$.

Problem 6

Hint 1. Decompose into partial fractions: $\dfrac{1}{(4k+1)(4k+2)(4k+3)(4k+4)} = \dfrac{1}{6}\Big(\dfrac{1}{4k+1} - \dfrac{3}{4k+2} + \dfrac{3}{4k+3} - \dfrac{1}{4k+4}\Big)$.

Hint 2. Write $\frac{1}{4k+j} = \int_0^1 x^{4k+j-1}\,dx$ and sum the geometric series (justify swapping, e.g. by monotone convergence on $[0,1)$): the sum is $\frac{1}{6}\int_0^1 \frac{(1-x)^3}{1-x^4}\,dx = \frac{1}{6}\int_0^1 \frac{(1-x)^2}{(1+x)(1+x^2)}\,dx$.

Hint 3. $\frac{(1-x)^2}{(1+x)(1+x^2)} = \frac{2}{1+x} - \frac{x+1}{1+x^2}$, giving the answer $\dfrac{\ln 2}{4} - \dfrac{\pi}{24}$.

Problem 7

Hint 1. The map $x \mapsto x^2 - 2$ is $t + t^{-1} \mapsto t^2 + t^{-2}$. Write $x_1 = \sqrt{5} = t + t^{-1}$.

Hint 2. $t = \frac{1+\sqrt{5}}{2}$, so $x_n = t^{2^{n-1}} + t^{-2^{n-1}}$, and the product telescopes: $(t - t^{-1})x_1 x_2 \cdots x_n = t^{2^n} - t^{-2^n}$.

Hint 3. Hence the ratio equals $\dfrac{1}{t - t^{-1}} \cdot \dfrac{t^{2^n} - t^{-2^n}}{t^{2^n} + t^{-2^n}} \to \dfrac{1}{t - t^{-1}} = 1$.

Problem 8

Hint 1. The sum is the "odd part" of a binomial expansion: $S_n = \sum_k \binom{n}{2k+1} 5^k = \dfrac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{2\sqrt{5}}$.

Hint 2. By Theorem 2, S_n satisfies $S_{n+1} = 2S_n + 4S_{n-1}$ ($1 \pm \sqrt{5}$ are the roots of $x^2 - 2x - 4$), with $S_1 = 1$, $S_2 = 2$. Prove $2^{n-1} \mid S_n$ by strong induction. (Equivalently, $S_n = 2^{n-1}F_n$.)

Problem 9

Note. The original IMC 2022 problem asks for the number of ways to be back at 0 after $p-1$ minutes, modulo p. The hints first treat the printed version (p minutes), and the last hint treats the original one.

Hint 1. The number of ways is the constant term of $(x^{-1} + 1 + x)^p$, i.e. $T_p = \sum_k \dfrac{p!}{k!\,k!\,(p-2k)!}$ (k left jumps, k right jumps).

Hint 2. Modulo p: $(x^{-1} + 1 + x)^p \equiv x^{-p} + 1 + x^p$ (Frobenius), whose constant term is 1; so $T_p \equiv 1 \pmod p$. Combinatorially: the cyclic shift acts on the walks, and every orbit has size p except the walk that always stays.

Hint 3. Original version: $f(p)=\sum_k \binom{p-1}{2k}\binom{2k}{k} \equiv \sum_{k=0}^{(p-1)/2}\binom{2k}{k} \pmod p$, since $\binom{p-1}{j}\equiv(-1)^j$. With $m=\frac{p-1}{2}\equiv -\frac12$ one has $\binom{2k}{k}\equiv(-4)^k\binom{m}{k}$, so the sum is $\equiv(1-4)^m=(-3)^{(p-1)/2}$, which by Euler's criterion and $\left(\frac{-3}{p}\right)=\left(\frac{p}{3}\right)$ is $\equiv 0,1,-1$ according as $p=3$, $p\equiv 1$, $p\equiv 2 \pmod 3$ (check: $f(3)=3$, $f(5)=19$, $f(7)=141$).

Problem 10

Hint 1. Get rid of the root: $(a_{n+2}-5(a_{n+1}+a_n))^2 = 4(6a_{n+1}^2+15a_{n+1}a_n+6a_n^2)$. Expand and look at the symmetry of the result.

Hint 2. It becomes $a_{n+2}^2+a_{n+1}^2+a_n^2 = 10(a_{n+2}a_{n+1}+a_{n+1}a_n+a_na_{n+2})$, symmetric in the three consecutive terms. So for $n\geqslant 2$ both a_{n+2} and a_{n-1} are roots of $z^2-10(a_{n+1}+a_n)z+(a_{n+1}^2+a_n^2-10a_{n+1}a_n)=0$.

Hint 3. The sequence is increasing, so $a_{n-1}<5(a_{n+1}+a_n)\leqslant a_{n+2}$: they are the two different roots, and Vieta gives $a_{n+2}=10(a_{n+1}+a_n)-a_{n-1}$. With $a_3=71$ computed directly, integrality follows by induction.

Problem 11

Note. Part (c) asks whether $\sum_{n\geqslant 0}\frac{2}{3^n+1}$ is rational. It is irrational, but the only known proofs (P. Borwein, *On the irrationality of* $\sum 1/(q^n+r)$, 1991) are far beyond olympiad level, so probably the intended question is different; the hint for (c) only records the answer.

(a)

Hint 1. Compute: $1,2,5,14,41,122,\ldots$; guess $a_n=\frac{3^n+1}{2}$.

Hint 2. Verify by induction: $\frac{3^{n+1}+1}{2}\cdot\frac{3^{n-1}+1}{2}=3^{n-1}+\left(\frac{3^n+1}{2}\right)^2$ (both sides minus $\frac{3^{2n}+1}{4}$ equal $\frac{10\cdot 3^{n-1}}{4}$). Equivalently $a_{n+1}=4a_n-3a_{n-1}$, which gives integrality directly.

(b)

Hint 1. $\frac{1}{a_n}=\frac{2}{3^n+1}<2\cdot 3^{-n}$: compare with a geometric series. It converges.

(c)

Hint 1. The sum is $\sum_{n\geqslant 0}\frac{2}{3^n+1}$, which is irrational by Borwein's theorem (for integer $q\geqslant 2$ and rational $r\neq 0$, $r\neq -q^n$, the sum $\sum 1/(q^n+r)$ is irrational).